THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 24	Height splits the base: $h^2 = 30^2 - 18^2 = 576$
2	**(d)** 84	Other side $= 24$; perimeter $= 2(18 + 24)$
3	**(b)** 48	Half of other diagonal $= \sqrt{30^2 - 18^2} = 24$
4	**(d)** 30	Height difference 18 m; $\sqrt{18^2 + 24^2} = 30$
5	**(b)** 216	Other leg $= 24$; area $= \frac{1}{2}ab$
6	**(a)** 33, 44, 55	Multiples of $(3, 4, 5)$ by 11 stay a triplet
7	**(c)** 36	Height splits the base: $h^2 = 39^2 - 15^2 = 1296$
8	**(b)** 102	Other side $= 36$; perimeter $= 2(15 + 36)$
9	**(d)** 72	Half of other diagonal $= \sqrt{39^2 - 15^2} = 36$
10	**(d)** 39	Height difference 15 m; $\sqrt{15^2 + 36^2} = 39$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Height bisects the base, $h^2 = 30^2 - 18^2$ (1); $h = 24$ cm (1).	1+1
12	Other side $= 32$ cm (1); area $= 768$ cm² (1).	1+1
13	Half of other diagonal$^2 = 50^2 - 30^2 = 1600$ (1); other diagonal $= 80$ cm (1).	1+1
14	Height bisects the base, $h^2 = 39^2 - 15^2$ (1); $h = 36$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Initial foot distance $= \sqrt{25^2 - 24^2} = 7$ m (1); new height $= \sqrt{25^2 - 15^2} = 20$ m (1); slip $= 24 - 20 = 4$ m (1).	1+1+1
16	$18^2 + 24^2 = 900 = 30^2$ (1); right-angled with legs 18, 24 (1); area $= \frac{1}{2} \times 18 \times 24 = 216$ cm² (1).	1+1+1
17	Hypotenuse $= 30$ cm (1); area $= \frac{1}{2} \times 18 \times 24 = 216$ (1); $\frac{1}{2} \times 30 \times h = 216$ gives $h = 14.4$ cm (1).	1+1+1
	OR: Initial foot distance $= \sqrt{25^2 - 20^2} = 15$ m (1); new height $= \sqrt{25^2 - 20^2} = 15$ m (1); slip $= 20 - 15 = 5$ m (1).	1+1+1
18	$15^2 + 36^2 = 1521 = 39^2$ (1); right-angled with legs 15, 36 (1); area $= \frac{1}{2} \times 15 \times 36 = 270$ cm² (1).	1+1+1
	OR: Hypotenuse $= 39$ cm (1); area $= \frac{1}{2} \times 15 \times 36 = 270$ (1); $\frac{1}{2} \times 39 \times h = 270$ gives $h = 13.85$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\sqrt{18^2 + 24^2} = 30$ m (2); (ii) saved $= 18 + 24 - 30 = 12$ m (1); (iii) total $= 18 + 24 + 30 = 72$ m (2).	2+1+2
	OR: Height$^2 = 40^2 - 24^2 = 1024$, height $= 32$ cm (2); area $= \frac{1}{2} \times 48 \times 32 = 768$ cm² (2); perimeter $= 2 \times 40 + 48 = 128$ cm (1).	2+2+1
20	(i) $\sqrt{15^2 + 36^2} = 39$ m (2); (ii) saved $= 15 + 36 - 39 = 12$ m (1); (iii) total $= 15 + 36 + 39 = 90$ m (2).	2+1+2
	OR: Height$^2 = 35^2 - 21^2 = 784$, height $= 28$ cm (2); area $= \frac{1}{2} \times 42 \times 28 = 588$ cm² (2); perimeter $= 2 \times 35 + 42 = 112$ cm (1).	2+2+1

Mistakes to avoid:

- Applying $c^2 = a^2 + b^2$ with the wrong side as hypotenuse; c is always the side opposite the right angle (the longest).
- Forgetting to take the square root at the end (giving 169 instead of 13).
- Adding the squares when finding a leg: leg $= \sqrt{c^2 - a^2}$ (subtract).
- Using the converse with the wrong pairing: always square the **longest** side on one side of the equation.
- Writing $\sqrt{a^2 + b^2} = a + b$; for example $\sqrt{9 + 16} = 5$, not $3 + 4 = 7$.
- Assuming three consecutive numbers such as 2, 3, 4 form a right triangle; always verify $a^2 + b^2 = c^2$ (here $4 + 9 \neq 16$).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 270	Other leg $= 36$; area $= \frac{1}{2}ab$
2	**(d)** 35, 84, 91	Multiples of $(5, 12, 13)$ by 7 stay a triplet
3	**(b)** 32	Height splits the base: $h^2 = 40^2 - 24^2 = 1024$
4	**(c)** 112	Other side $= 32$; perimeter $= 2(24 + 32)$
5	**(a)** 64	Half of other diagonal $= \sqrt{40^2 - 24^2} = 32$
6	**(d)** 40	Height difference 24 m; $\sqrt{24^2 + 32^2} = 40$
7	**(a)** 384	Other leg $= 32$; area $= \frac{1}{2}ab$
8	**(a)** 40, 75, 85	Multiples of $(8, 15, 17)$ by 5 stay a triplet
9	**(d)** 28	Height splits the base: $h^2 = 35^2 - 21^2 = 784$
10	**(a)** 98	Other side $= 28$; perimeter $= 2(21 + 28)$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Other side $= 28$ cm (1); area $= 588$ cm² (1).	1+1
12	Half of other diagonal$^2 = 50^2 - 14^2 = 2304$ (1); other diagonal $= 96$ cm (1).	1+1
13	Height bisects the base, $h^2 = 40^2 - 24^2$ (1); $h = 32$ cm (1).	1+1
14	Other side $= 40$ cm (1); area $= 1200$ cm² (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Initial foot distance $= \sqrt{13^2 - 12^2} = 5$ m (1); new height $= \sqrt{13^2 - 12^2} = 5$ m (1); slip $= 12 - 5 = 7$ m (1).	1+1+1
16	$24^2 + 32^2 = 1600 = 40^2$ (1); right-angled with legs 24, 32 (1); area $= \frac{1}{2} \times 24 \times 32 = 384$ cm² (1).	1+1+1
17	Hypotenuse $= 40$ cm (1); area $= \frac{1}{2} \times 24 \times 32 = 384$ (1); $\frac{1}{2} \times 40 \times h = 384$ gives $h = 19.2$ cm (1).	1+1+1
	OR: Initial foot distance $= \sqrt{17^2 - 15^2} = 8$ m (1); new height $= \sqrt{17^2 - 15^2} = 8$ m (1); slip $= 15 - 8 = 7$ m (1).	1+1+1
18	$21^2 + 28^2 = 1225 = 35^2$ (1); right-angled with legs 21, 28 (1); area $= \frac{1}{2} \times 21 \times 28 = 294$ cm² (1).	1+1+1
	OR: Hypotenuse $= 35$ cm (1); area $= \frac{1}{2} \times 21 \times 28 = 294$ (1); $\frac{1}{2} \times 35 \times h = 294$ gives $h = 16.8$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\sqrt{24^2 + 32^2} = 40$ m (2); (ii) saved $= 24 + 32 - 40 = 16$ m (1); (iii) total $= 24 + 32 + 40 = 96$ m (2).	2+1+2
	OR: Height$^2 = 50^2 - 30^2 = 1600$, height $= 40$ cm (2); area $= \frac{1}{2} \times 60 \times 40 = 1200$ cm² (2); perimeter $= 2 \times 50 + 60 = 160$ cm (1).	2+2+1
20	(i) $\sqrt{21^2 + 28^2} = 35$ m (2); (ii) saved $= 21 + 28 - 35 = 14$ m (1); (iii) total $= 21 + 28 + 35 = 84$ m (2).	2+1+2
	OR: Height$^2 = 50^2 - 14^2 = 2304$, height $= 48$ cm (2); area $= \frac{1}{2} \times 28 \times 48 = 672$ cm² (2); perimeter $= 2 \times 50 + 28 = 128$ cm (1).	2+2+1

Mistakes to avoid:

- Applying $c^2 = a^2 + b^2$ with the wrong side as hypotenuse; c is always the side opposite the right angle (the longest).
- Forgetting to take the square root at the end (giving 169 instead of 13).
- Adding the squares when finding a leg: leg $= \sqrt{c^2 - a^2}$ (subtract).
- Using the converse with the wrong pairing: always square the **longest** side on one side of the equation.
- Writing $\sqrt{a^2 + b^2} = a + b$; for example $\sqrt{9 + 16} = 5$, not $3 + 4 = 7$.
- Assuming three consecutive numbers such as 2, 3, 4 form a right triangle; always verify $a^2 + b^2 = c^2$ (here $4 + 9 \neq 16$).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 56	Half of other diagonal $= \sqrt{35^2 - 21^2} = 28$
2	**(d)** 35	Height difference 21 m; $\sqrt{21^2 + 28^2} = 35$
3	**(c)** 294	Other leg $= 28$; area $= \frac{1}{2}ab$
4	**(d)** 21, 72, 75	Multiples of $(7, 24, 25)$ by 3 stay a triplet
5	**(d)** 40	Height splits the base: $h^2 = 50^2 - 30^2 = 1600$
6	**(b)** 140	Other side $= 40$; perimeter $= 2(30 + 40)$
7	**(a)** 80	Half of other diagonal $= \sqrt{50^2 - 30^2} = 40$
8	**(b)** 50	Height difference 30 m; $\sqrt{30^2 + 40^2} = 50$
9	**(c)** 600	Other leg $= 40$; area $= \frac{1}{2}ab$
10	**(d)** 39, 52, 65	Multiples of $(3, 4, 5)$ by 13 stay a triplet

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Half of other diagonal$^2 = 45^2 - 27^2 = 1296$ (1); other diagonal $= 72$ cm (1).	1+1
12	Height bisects the base, $h^2 = 35^2 - 21^2$ (1); $h = 28$ cm (1).	1+1
13	Other side $= 48$ cm (1); area $= 672$ cm² (1).	1+1
14	Half of other diagonal$^2 = 58^2 - 40^2 = 1764$ (1); other diagonal $= 84$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Initial foot distance $= \sqrt{26^2 - 24^2} = 10$ m (1); new height $= \sqrt{26^2 - 24^2} = 10$ m (1); slip $= 24 - 10 = 14$ m (1).	1+1+1
16	$30^2 + 40^2 = 2500 = 50^2$ (1); right-angled with legs 30, 40 (1); area $= \frac{1}{2} \times 30 \times 40 = 600$ cm² (1).	1+1+1
17	Hypotenuse $= 50$ cm (1); area $= \frac{1}{2} \times 30 \times 40 = 600$ (1); $\frac{1}{2} \times 50 \times h = 600$ gives $h = 24$ cm (1).	1+1+1
	OR: Initial foot distance $= \sqrt{39^2 - 36^2} = 15$ m (1); new height $= \sqrt{39^2 - 36^2} = 15$ m (1); slip $= 36 - 15 = 21$ m (1).	1+1+1
18	$14^2 + 48^2 = 2500 = 50^2$ (1); right-angled with legs 14, 48 (1); area $= \frac{1}{2} \times 14 \times 48 = 336$ cm² (1).	1+1+1
	OR: Hypotenuse $= 50$ cm (1); area $= \frac{1}{2} \times 14 \times 48 = 336$ (1); $\frac{1}{2} \times 50 \times h = 336$ gives $h = 13.44$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\sqrt{30^2 + 40^2} = 50$ m (2); (ii) saved $= 30 + 40 - 50 = 20$ m (1); (iii) total $= 30 + 40 + 50 = 120$ m (2).	2+1+2
	OR: Height$^2 = 45^2 - 27^2 = 1296$, height $= 36$ cm (2); area $= \frac{1}{2} \times 54 \times 36 = 972$ cm² (2); perimeter $= 2 \times 45 + 54 = 144$ cm (1).	2+2+1
20	(i) $\sqrt{14^2 + 48^2} = 50$ m (2); (ii) saved $= 14 + 48 - 50 = 12$ m (1); (iii) total $= 14 + 48 + 50 = 112$ m (2).	2+1+2
	OR: Height$^2 = 58^2 - 40^2 = 1764$, height $= 42$ cm (2); area $= \frac{1}{2} \times 80 \times 42 = 1680$ cm² (2); perimeter $= 2 \times 58 + 80 = 196$ cm (1).	2+2+1

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- Adding the squares when finding a leg: leg $= \sqrt{c^2 - a^2}$ (subtract).
- Using the converse with the wrong pairing: always square the **longest** side on one side of the equation.
- Writing $\sqrt{a^2 + b^2} = a + b$; for example $\sqrt{9 + 16} = 5$, not $3 + 4 = 7$.
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